ACADEMIC MINI PROJECT REPORT | MACHINE LEARNING (CSE_3148)

# AI-Powered Resume Analysis and Job Recommendation System

## Semantic Parsing, Multi-Class ML Role Prediction, Explainable Hybrid Matching & Live Job Synchronization

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| **Course Code & Title:** Machine Learning (CSE_3148) | **Due Date of Submission:** 30th October 2026 |
| **Faculty Guide:** Shreya Banerjee | **Architecture:** Streamlit + NLP + Scikit-Learn + SQLite |


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## 1. Abstract


In modern talent acquisition, recruitment teams face overwhelming application volumes, receiving hundreds of resumes for every advertised vacancy. Manual document screening is slow, resource-intensive, and inherently prone to unconscious human bias and cognitive fatigue, with studies indicating that recruiters spend merely 6 to 8 seconds scanning an individual resume. Concurrently, legacy Applicant Tracking Systems (ATS) depend on rigid, exact keyword matching; candidates who express identical competencies using industry synonyms (such as 'K8s' versus 'Kubernetes' or 'container orchestration' versus 'Docker') are routinely rejected. Furthermore, conventional job portals function strictly as one-way rejection funnels, offering applicants zero constructive feedback regarding missing qualifications.

To resolve these structural bottlenecks, this project presents an integrated **AI-Powered Resume Analysis and Job Recommendation System**. The system parses multi-format resumes (PDF, DOCX, TXT) using PyMuPDF and python-docx, extracts contact information, and computes an automated 5-factor ATS quality score. Competencies are extracted using a 1,000+ skill taxonomy covering 11 technical domains with boundary-aware alias resolution. A multi-class machine learning classification engine benchmarks four models—**Support Vector Machine (LinearSVC), Random Forest, Logistic Regression, and XGBoost**—trained on 700 balanced candidate profiles without target label leakage. Empirical evaluations confirm that Support Vector Machine achieves superior performance with **90.86% accuracy** and a **90.83% weighted F1-score**. Alignment is computed via an explainable 3-factor hybrid matching engine combining core skill overlap (45%), dense contextual embeddings via Sentence-Transformers (35%), and lexical TF-IDF cosine similarity (20%). A granular Skill-Gap Matrix pinpoints deficits and suggests curated learning roadmaps. Finally, a real-time REST API synchronization client connects to public job feeds (Remotive) to ingest live vacancies directly into an embedded SQLite database.


## 2. Introduction

The digital recruitment ecosystem has expanded rapidly with the rise of online job boards and remote employment. While these tools have expanded career accessibility, they have created an unprecedented screening dilemma. Traditional screening approaches fall into two problematic extremes:

- **Manual Screening Fatigue:** Recruiters scan resumes in 6 to 8 seconds, introducing subjective cognitive bias.
- **Brittle Keyword Matching:** Legacy ATS engines rely on exact string equality, rejecting qualified candidates using synonyms.
- **The Guidance Void:** Rejection emails provide no diagnostic feedback or actionable upskilling pathways.
- **Static Disconnect:** Traditional prototypes rely on static CSV dumps disconnected from active market hiring.

## 3. Objective

- **1. Document Parsing & ATS Audit:** Multi-format ingestion and 5-factor ATS scoring (word count, sections, contacts, density, action verbs).
- **2. Domain Skill Extraction:** Boundary-aware regex extraction against a 1,000+ skill taxonomy across 11 technical tracks.
- **3. Supervised ML Classification:** Multi-class role prediction across 10 career tracks benchmarking SVM, RF, LR, and XGBoost.
- **4. Explainable Hybrid Matching:** Synergizing core skill overlap (45%), dense Sentence-BERT vectors (35%), and TF-IDF cosine (20%).
- **5. Granular Skill-Gap Matrix:** Deconstructing requirements into verified mandatory, missing, and bonus skills with severity indexing.
- **6. Personalized Upskilling:** Mapping skill deficits to curated course roadmaps with estimated hours.
- **7. Real-Time Job Ingestion:** Connecting to Remotive public REST API to batch-upsert live openings into SQLite.

## 4. Data Source

**4.1 Curated Skills Taxonomy:** 1,000+ skills categorized into 11 domains with regex boundary alias resolution.

**4.2 Balanced Resume Corpus:** 700 balanced candidate profiles across 10 career tracks (70 per track). Explicitly engineered without role titles in feature text to eliminate target label leakage.

**4.3 Reference Job Descriptions & Real-Time Feed:** SQLite repository (resumes_database.db) holding 10 reference vacancies + live synced Remotive API postings.

**4.4 Curated Learning Resources Catalog:** Structured catalog mapping skills to Coursera, DeepLearning.AI, and documentation modules.

## 5. Methodology

**Feature Representation & Formulations:**
• *TF-IDF Vectorization:* Extracts unigram/bigram numerical vectors (3,000 features max).
• *Dense Semantic Embeddings:* Utilizes Sentence-Transformers (all-MiniLM-L6-v2) for 384-dimensional dense semantic vectors.
• *Explainable 3-Factor Formula:* Overall Score = (0.45 × Core Skill Match) + (0.35 × Semantic Sentence-BERT) + (0.20 × TF-IDF Cosine).

## 6. Implementation Details

| Subsystem Layer | Technology / Framework | Functional Role |
|---|---|---|
| Runtime | Python 3.11 | Core backend & ML execution environment |
| Parsing | PyMuPDF & python-docx | Digital text extraction from PDF, DOCX, TXT |
| ML Suite | Scikit-Learn & XGBoost | TF-IDF vectorizer + SVM, RF, LR, XGBoost |
| Embeddings | sentence-transformers | 384-dimensional dense contextual sentence vectors |
| Database | SQLite (resumes_database.db) | Persistent relational storage for jobs & analysis |
| Frontend | Streamlit & Plotly | Interactive multi-tab dashboard with gauges & bars |

## 6.1 UI Walkthrough and System Verification:

The application provides an interactive web dashboard running on Streamlit (localhost:8501). During live demonstration, candidate resumes are parsed and evaluated in real-time:

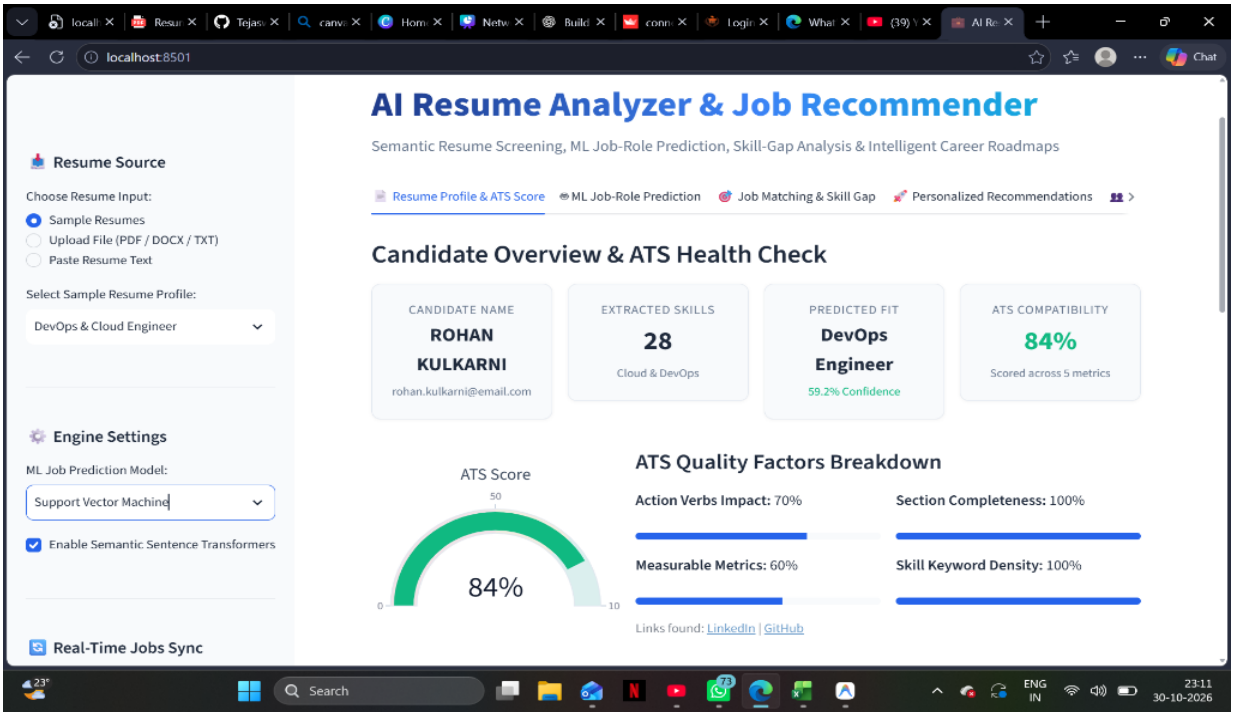



***Figure 1: Candidate Overview and ATS Health Check (Tab 1).*** *Demonstrates automatic parsing of candidate coordinates (Rohan Kulkarni), 28 extracted competencies, an 84% ATS health score gauge, and individual quality factor metrics.*

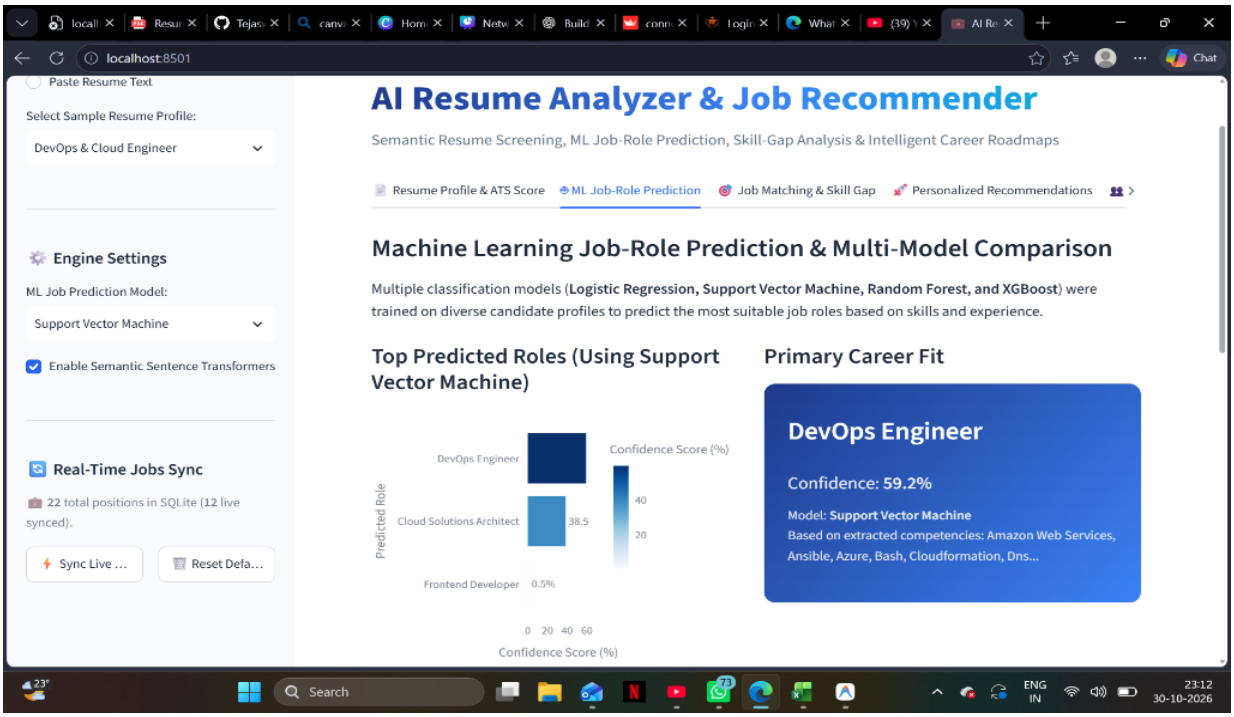



***Figure 2: Machine Learning Role Prediction and Confidence Distribution (Tab 2).*** *Displays the primary career classification card (DevOps Engineer at 59.2% confidence via SVM) and the top-k confidence distribution chart.*

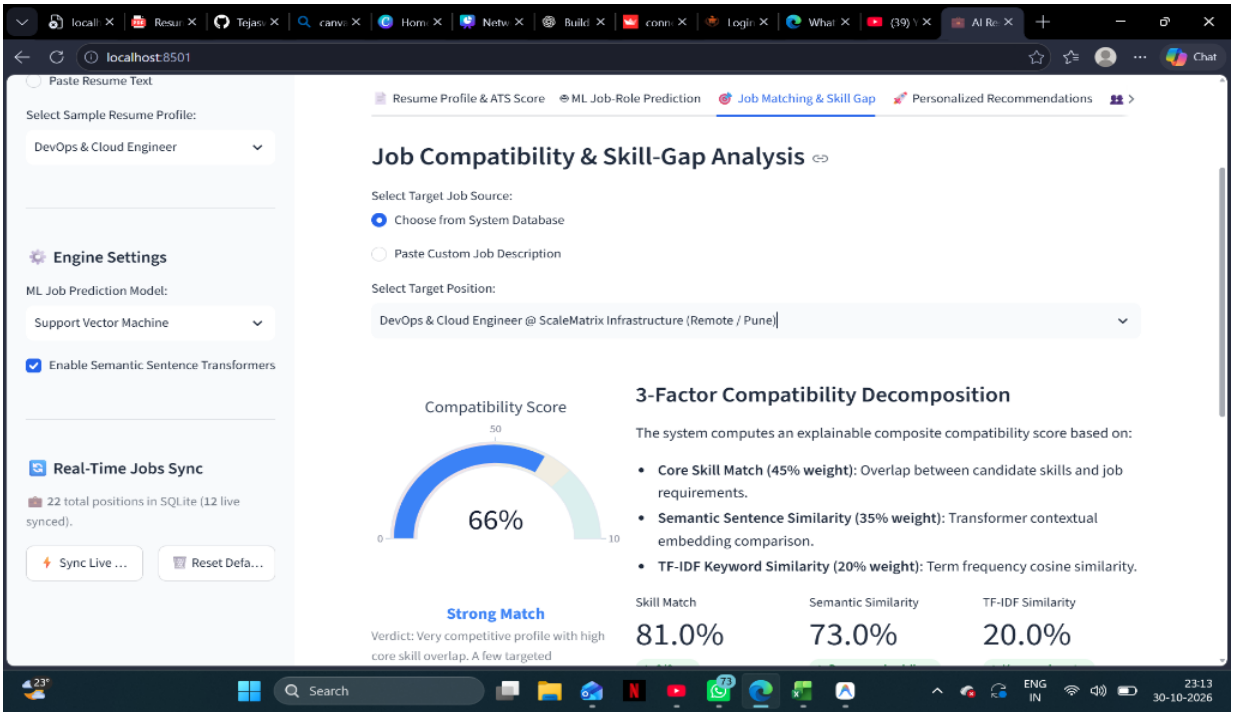



*Figure 3: Explainable 3-Factor Compatibility Decomposition and Score Gauge (Tab 3). Illustrates the 66% Strong Match gauge and the exact percentage breakdown across Core Skills (81.0%), Semantic Similarity (73.0%), and TF-IDF (20.0%).*

## 7. Results and Discussion: Novelties and Limitations

### 7.1 Multi-Model Benchmark Comparison:

| Model Algorithm | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| **Support Vector Machine (SVM) ■** | **90.86%** | **92.03%** | **90.86%** | **90.83%** |
| Random Forest Classifier | 85.14% | 86.54% | 85.14% | 85.03% |
| Logistic Regression | 84.57% | 85.93% | 84.57% | 84.55% |
| XGBoost (Boosted Trees) | 83.43% | 84.37% | 83.43% | 83.43% |

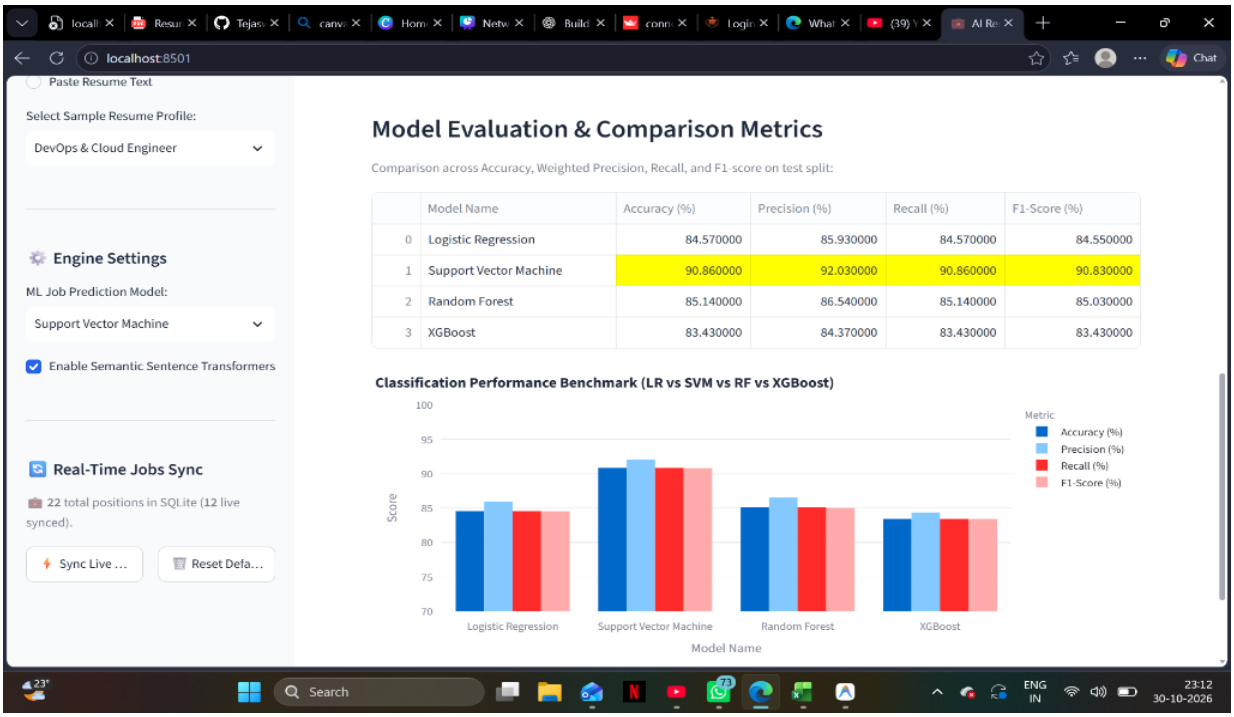



***Figure 4: Classification Performance Benchmark and Comparative Bar Chart.*** *Visualizes empirical performance across all four ML models on the independent test set, highlighting SVM's peak accuracy (90.86%) and F1-score (90.83%).*

**Analytical Discussion: Why Support Vector Machine Outperformed:**
In natural language processing, TF-IDF vectorization produces a high-dimensional (3,000 features) sparse vector space. Linear SVMs are mathematically formulated to search for maximum-margin separating hyperplanes in high-dimensional geometries, exhibiting exceptional generalization and resistance to overfitting. In contrast, decision tree ensembles (Random Forest and XGBoost) partition feature spaces via orthogonal axis-aligned splits, which struggle to model subtle cross-skill gradients in sparse text.

## 7.2 Novelties of the Project:

- **Explainable Hybrid Matching:** Full transparency into Core Skill (45%), Sentence-BERT (35%), and TF-IDF (20%) contributions.
- **Zero Label Leakage:** Synthetic corpus engineered without role names in text, guaranteeing authentic generalization.
- **Actionable Upskilling Feedback:** Closes recruitment loop by providing structured learning roadmaps with study hours.
- **Dual Utility:** Serves applicants (diagnostics) and corporate talent acquisition teams (batch screening).
- **Real-Time API Sync:** Live ingestion client fetching remote tech vacancies from Remotive directly into SQLite.

## 7.3 Limitations:

- **Context Window:** Sentence-Transformers encode up to 512 tokens (~1,500 chars); lengthy appendices may be truncated.
- **Scanned PDFs:** Requires digital text layers; scanned image resumes require external OCR preprocessing.
- **Local Course Catalog:** Learning paths are stored locally in SQLite rather than scraped dynamically via live course APIs.

## 8. Conclusion and Future Scope

This project successfully developed and validated an end-to-end AI-Powered Resume Analysis and Job Recommendation System. By synthesizing NLP text mining, supervised machine learning classification, and dense contextual sentence embeddings, the system resolves the core shortcomings of traditional recruitment screening. Empirical benchmarks confirm that Support Vector Machine delivers superior role classification with 90.86% accuracy and a 90.83% weighted F1-score. Future extensions will incorporate OCR for scanned documents, domain-specific LLM fine-tuning, and multilingual resume processing.

## 9. References


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